

RESEARCH ARTICLE

Quantum Safe Proxy Blind Signature Protocol Based on 3D Entangled GHZ-Type States

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ABSTRACT

As quantum technology advances, classical digital signatures exhibit vulnerabilities in preserving security properties during the transmission of information. Working toward a reliable communication protocol, we introduce a proxy blind signature scheme to teleport a single particle qubit state with a message to the receiver, employing a three qubit GHZ entangled state. The blindness property is utilized to secure the message information from the proxy signer. A trusted party, Trent, is introduced to supervise the communication process. Alice blinds the original message and sends the Bell measurements with her entangled particle to proxy signer Charlie. After receiving measurements from Alice and Charlie, Bob verifies the proxy blind signature and performs appropriate unitary operations on his particle. Thereafter, Trent verifies the security of the quantum teleportation setup by matching the output data with the original data sent by Alice. Security analysis results prove that the proposed scheme fulfils the basic security necessities, including undeniability, unforgeability, blindness, verifiability, and traceability.

1 | Introduction

Quantum cryptography (QC) leverages the properties of quantum mechanics and traditional cryptography to establish a secure channel. With the onset of quantum computers and advancements in techniques to decode or intercept information, classical cryptography failed to provide optimal security. At this juncture, the genesis of QC occurred, marking the inception of quantum key distribution, a concept introduced by Bennett and Brassard in 1984 [1]. QC ensures unconditional security for transferring information by utilizing the laws of quantum mechanics, including Heisenberg's Uncertainty Principle, the no-cloning theorem, and no-deleting theorem. Due to the reliability that QC

provides, modern times have made research extremely important in this field. Various important branches of QC include QKD, Quantum Teleportation (QT) [2–4], Superdense Coding [5–7], Bidirectional Quantum Teleportation (BQT) [8, 9], and Quantum Signature (QS) [10–12].

QT is a technique for processing quantum information, enabling the transfer of an unknown state via the properties of entangled particle correlations and classical communication. QT only teleports the information of the quantum particle, not the particle itself. The basis of QT lies in different types of quantum measurements (Lee et al. [13] a theory of their interrelation) and operations that reverse the superposition state to its original

Abbreviations: AQS, arbitrated quantum signature; QBS, quantum blind signature; QC, quantum cryptography; QGS, quantum group signature; QKD, quantum key distribution; QMS, quantum multi-party signature; QPBS, quantum proxy blind signature; QPS, quantum proxy signature; QT, quantum teleportation.

state. Quantum entanglement is an important constituent of QT. The entanglement is a unique interconnection between quantum bits. The entangled quantum particles keep sharing information about states in between and make up the quantum channel. Quantum-superimposed particles are entangled to improve the security and reliability of the network. The main challenge of using QT is to ensure the security and reliability of the communication protocol. The noise and interference corrupt the quantum state and make it useless because the quantum state is usually sensitive to these during QT. Consequently, the complete security of the system can be achieved by using different types of signatures: arbitrated quantum signature (AQS) [14, 15], quantum proxy signature (QPS) [16, 17], quantum blind signature (QBS) [18, 19], quantum group signature (QGS) [20], quantum multi-party signature (QMS) [21–23], quantum multiproxy signature [24], quantum multiblind signature [25, 26], and quantum proxy blind signature (QPBS) [27–30].

There are a number of practical applications in real life where digital signatures are useful, such as e-commerce, e-banking, and e-voting. The digital signature is a method to verify the authenticity and integrity of a message. The loopholes in classical digital signatures are progressively increasing as quantum algorithms continue to advance further. Quantum signatures are employed to ensure the integrity and authenticity of our security scheme. A quantum signature (QS) is generally a quantum communication equivalent to the digital signature [31, 32] that utilizes unknown quantum states to enhance the security of our communication. Depending on the purpose, we can use different quantum signatures. Quantum proxy signatures are a special type of signature where the original signer provides another party with authority to sign the message on his behalf. Signing privileges allow the authorized signer to generate signatures on behalf of the original owner. Thus, QPS plays an important role when the original signer cannot sign the message himself due to some problem, such as workload, and he or she wants to assign the responsibility to someone else. Quantum Blind signatures use the blindness property to hide the original information of the message from the signing authority. QBS helps ensure maximum privacy for the original owner of the message by hiding his identity. The blind signer can generate the original signature, given that he or she does not have access to the original content of the message. The accuracy of the signer's ability to trace the identity of the message owner divides blind signatures into weak QBS and strong QBS. A quantum proxy blind signature gives the message owner the right to authorize another party (the proxy signer) to generate a QBS on his behalf. The QPBS is verified in the same way we verify a proxy signature. By integrating the features of QPS and QBS, we obtain a QPBS. This signature is quite useful in a number of real-life applications, including e-voting, e-banking, e-commerce, and e-payment systems.

We present a scheme by using QPBS on a three-particle GHZ-entangled state to enable and secure one-directional QT. This scheme is secured to guarantee the privacy of users, network security, communication security, and reliability of the network, among others. The proposed scheme is subjected to various venomous attacks to demonstrate its robustness. The attack strategies include intercept-resend attack, signature forgery, man-in-the-middle attack, service denial, repudiation, disavowability, entanglement, and measure attack. The resilience of the

proposed QPBS protocol against these attacks gives evidence for its robustness.

2 | Related Work

The science of cryptography made a significant advancement with the introduction of quantum key distribution (QKD). The first QKD protocol, known as BB84, was introduced by Bennett and Brassard in 1984 [1]. The sender and receiver agree on a secret shared key, and the eavesdropper is not able to retrieve the information. The classical bit sequence of the message is transmitted either on a rectilinear or diagonal basis. The impossibility of measuring the state at once in two conjugate bases ensures the secrecy of the shared key. Unlike traditional cryptography, QC keeps the data secret due to the characteristics of quantum mechanics rather than the difficulty of doing specific computations. Bennett et al. [33] was the first to put up the idea of QT in 1993. Alice made a Bell measurement (BM) on her EPR particle and the message's quantum state. Bob uses the output of this measurement to replicate a copy of the message's quantum state from his preshared EPR particle. QT has undergone constant improvement with the advancement of quantum technology. Greenberger, Horne, and Zeilinger [34] introduced a highly entangled quantum state of three or more qubits in 1989 and called it the GHZ state.

The digital signature was separately proposed by Diffie and Hellman [35] and Merkle [36]. The first QDS scheme based on a quantum one-way function was introduced by Gottesman et al. [37] in 2001. To teleport using the GHZ states, the AQS scheme was proposed by Zeng et al. [38] in 2002. A lot of quantum signature schemes have since been presented. QBS focuses on signing the blinded form of the original message in an effective way. One such protocol relying on QBS was put forth by Wen et al. [39] in 2009. The signer, Bob, is the representative of the bank. The receiver (merchant) can forge a valid signature by modifying the information. QPS is employed on a large scale owing to the flexibility it offers. Chen et al. [40] used EPR states to come up with a QPS scheme for QT in 2010. Since then, work on integrated quantum proxy and blind signatures has gained momentum. One such improvement scheme was introduced by Zhang et al. [41] with a six-qubit entangled state in 2018 where a trusted party Trent is brought in to avoid malicious activity from David. The signers can still be traced, even though they are blind to the content of the signed communication. In 2019, Chen et al. [42] used a nonorthogonal state of BB84 to come up with a signature scheme. The scheme added a randomizing factor to the message and removed the blind factor from QBS. Xia et al. [43] suggested a semi-QBS mechanism in 2021 utilizing the five qubit GHZ state to enhance the security during the exchange of information between the parties and employing a new eavesdropping detection technique. Dianjun et al. [44] presented a verifiable AQS scheme using five qubit entangled states as a quantum channel based on controlled QT. The scheme used mutually unbiased base particles and symmetric bivariate polynomials. Qiming et al. [45] suggested a protocol in 2022 to exchange two unknown states with messages among the authorized users by employing an entangled state of five particles. Zhang et al. [46] investigated QT by using quantum channels of GHZ state and W state within a noisy environment in 2023. Men et al. [47] presented

a scheme in 2023 and used a quantum Fourier transform (QFT) as a method to encrypt the message, improving the quantum efficiency. Tabatabaei et al. [48] gave a method in 2023 to bidirectionally transmit n -qubit states at the same time between two users. Recent years have seen significant advancements in the use of quantum signature schemes for secure communication, leveraging multiparticle entangled states for QT.

2.1 | Motivation and Contributions

The complexities and security issues in previously given signatures, whether classical signatures, semiquantum signatures, or quantum signatures, motivated us to work on an efficient quantum signature. The security properties of different signatures are integrated into the proposed scheme to design a quantum proxy blind signature. This signature offers enhanced security by using the correlation of entangled particles and eliminating the need for signers to reveal content. This is particularly important for sensitive information like medical records or financial documents. This signature can be incorporated into quantum networks, providing a robust and scalable security solution. Further exploring proxy signatures in QT could lead to theoretical advancements in cryptography and quantum information science and practical applications like secure e-commerce and e-voting. Below, we outline our contributions:

1. Initially, we propose a quantum proxy-blind signature protocol utilizing a trusted party. This protocol capitalizes on a three-qubit GHZ-entangled state for QT, aiming for a reduced error rate and expedited information transfer. The security of the communication process is ensured through the application of blindness properties, incorporating hash functions, XOR operations, and a specific rule for acquiring quantum states.
2. Subsequently, we subject the signature scheme to various security challenges, encompassing inter-resend attacks, man-in-the-middle attacks, and entanglement measure attacks. Our scheme shows security properties like unforgeability, blindness, traceability, and nonrepudiation. The security analysis underscores the robustness of our proposed scheme, grounded in the principles of the quantum realm.
3. At last, we use the “Python” and “SCYTHY” tools to showcase the performance of our proposed protocol and analyze its behavior during the communication process. The experimental analysis substantiates the adaptability of our protocol across diverse environments.

3 | Preliminaries

3.1 | GHZ State

Our proposed scheme works on the entangled GHZ state. The three-qubit GHZ state is a superposition of $|000\rangle$ and $|111\rangle$, and the qubits only acquire definite values upon measurement. This state displays a high level of entanglement due to the entanglement between the state of each qubit and the states of the other two qubits. Once the first qubit is measured and found in the state

$|0\rangle$, then the remaining two qubits will collapse to the state $|0\rangle$ as well. Similarly, if the first qubit is found in $|1\rangle$, the remaining qubits will collapse to $|1\rangle$. This characteristic does not change regardless of the physical distance between the qubits. The GHZ state has been used to violate Bell’s inequality [49]. The GHZ state highlights the attractive attributes of quantum mechanics and has the potential to transform our knowledge of the universe.

3.2 | Properties of the QPBS

The signing protocol for QPBS consists of five steps: blinding the message, authorizing the proxy signer, blind signing the message, verifying the signature, and unblinding and verifying the message. After performing the blind transformation on the original message, denoted as m , Alice acquires the blinded message M . Subsequently, Alice proceeds to transmit the altered blinded data M to Charlie, who serves as the proxy signer. Charlie, in turn, adds his signature on M , resulting in the blind signature $\text{Sign}(M)$. This blind signature is subsequently conveyed to the message verifier, the trusted party, and the recipient of the message. Trent securely obtains the communication from Bob. Trent then proceeds to eliminate the blindness from $\text{Sign}(M)$ and retrieve the signature $\text{Sign}(m)$ corresponding to the initial message m . Bob, the individual responsible for verifying signatures, possesses the ability to determine the accuracy of $\text{Sign}(m)$. Lastly, an ideal proxy blind signature is expected to encompass the following attributes:

- **Unforgeability:** A verifiable proxy blind signature can only be created by the signer.
- **Undeniability:** Once a sender has signed a message or a receiver has received a message, they cannot repudiate their signature on the message.
- **Blindness:** The message owner has the information of the original message, and his blinding of the message prevents the signer from knowing the original contents.
- **Verifiability:** The signature and the message can both be confirmed without any issues, even if the message is blinded or signed by a proxy signer.
- **Identifiability:** To resolve any dispute, QPBS can be used to find out the proxy signer’s identity.

Proxy-blind signatures satisfying these properties are considered secure. These are the criteria we need to follow when we design a QPBS;

3.3 | Controlled QT

The GHZ state has been utilized to execute QT, a technique that facilitates the transfer of a quantum state without physically transmitting the particle. QT hinges on the principle of entanglement, and the GHZ state is especially advantageous in this context. Consider a three-qubit GHZ state involving the sender (Alice), receiver (Bob), and controller (Charlie):

$$|GHZ\rangle_{123} = \frac{1}{\sqrt{2}}(|000\rangle_{123} + |111\rangle_{123}) \quad (1)$$

TABLE 1 | Notation table.

Serial No.	Notation	Description
1	μ	Bit sequence of original message
2	M	Blinded message
3	h	Hash function
4	S	Signature
5	Ω_A	Measurements recorded by Alice
6	Ω_C	Measurements recorded by Charlie
7	K_{TA}	Shared key of Trent and Alice
8	K_{TB}	Shared key of Trent and Bob
9	K_{TC}	Shared key of Trent and Charlie
10	K_{AB}	Shared key of Alice and Bob
11	K_{CB}	Shared key of Charlie and Bob
12	K_{CA}	Shared key of Charlie and Alice
13	$ \rangle$	Ket (representation of a Qubit)
14	$\oplus$	XOR operation

where particle “1” is with Alice, particle “2” is with Bob, and particle “3” is with Charlie. Additionally, Alice possesses a particle in a blinded state μ given by

$$|\phi\rangle_M = \frac{1}{\sqrt{2}}(|0\rangle + b|1\rangle)_M \quad (2)$$

where $b = 1$ or $b = -1$ depending on $\mu(i) = 0$ or $\mu(i) = 1$ for $i = 1, 2, \dots, l$. The combined state of Alice’s blinded qubit and the GHZ state can be written as

$$|\phi\rangle_{M123} = |\phi\rangle_M \otimes |\phi\rangle_{123} \quad (3)$$

which leads to

$$|\phi\rangle_{M123} = \frac{1}{2}(|0000\rangle_{M123} + |0111\rangle_{M123} + b|1000\rangle_{M123} + b|1111\rangle_{M123}) \quad (4)$$

Next, Alice performs Bell measurements (B.M.s) on her qubits “(1,2).” The possible Bell states are given by

$$|\phi\rangle_{\pm} = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), |\phi\rangle_{\pm} = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle) \quad (5)$$

After Alice’s Bell measurements, Charlie performs a single-particle measurement on particle “3.” Depending on the outcomes of Alice’s Bell measurements and Charlie’s measurement, the state of particle “2” held by Bob is transformed into a state related to μ . Bob then applies a corresponding unitary operation to his particle “2” to recover the state of particle μ , effectively teleportation the state from Alice to Bob (Table 1).

4 | Our QPBS Scheme

The stakeholders in our security protocol are as follows: (i) Alice is the individual who owns the message. (ii) Bob assumes the role of the recipient and undertakes the responsibility of message verification. (iii) Charlie operates as the authorized proxy signer. (iv)

Trent, acting as a message validator, fulfills the role of a trusted party. A three-qubit GHZ-entangled state is used to enable QT between the various parties. Alice sends a blinded message to Charlie and authorizes him to do proxy signing in his place. Bob verifies the signature and Trent verifies the message. The comprehensive procedure of our technique can be laid out in the manner shown below:

4.1 | Initializing Phase

- **Quantum Key Distribution:** The QKD protocol is adopted to share the following keys: K_{AB} is shared by Bob and Alice; K_{TA} is shared by Alice and Trent; K_{TB} is shared by Bob and Trent; K_{CA} is shared by Alice and Charlie; K_{CB} is shared by Bob and Charlie; and K_{TC} is shared by Trent and Charlie.
- **Quantum Channel installation:** Bob generates l quantum states by using the GHZ state described in Equation (1).

$$|\phi\rangle_{123} = \frac{1}{\sqrt{2}}(|000\rangle_{123} + |111\rangle_{123}) \quad (6)$$

The entangled particles are now sent to different parties, l particles of (1) to Alice, l particles of (3) to Charlie and keeps to himself l particles of (2).

- Bob executes frequent eavesdropping checks (See example 1 and 2) for maintaining the reliability of the quantum channel.

Example 1. Bob and Alice perform eavesdropping detection by selecting a random subset of qubits and measuring them in randomly chosen complementary bases. They then compare the results over a classical channel to detect any inconsistencies. If the error rate exceeds a certain threshold, they abort the protocol.

Example 2. Bob executes eavesdropping checks after every transmission of n qubits, where n is a parameter that can be adjusted depending on the security level required.

4.2 | Blinding Phase

- Alice converts the message into an n -bit sequence such that $\mu = \{\mu(1), \mu(2), \dots, \mu(n)\}$ and transmits the generated sequence μ to Trent.
- Alice ensures that the transmission of μ occurs over a quantum-secured channel to prevent interception.
- Alice applies a hash function $h : \{0, 1\}^n \rightarrow \{0, 1\}^l (n \ll l)$ to convert μ into $h(\mu)$, sharing the details with Trent. Alice blinds μ by computing $M(i) = K_{TA} \oplus \mu(i)$ for $i = 1, \dots, l$, where $\oplus$ denotes the XOR operation. Alice then generates l quantum states based on the blinded information:

$$|\phi\rangle_{M(i)} = \frac{1}{\sqrt{2}}(|0\rangle + b|1\rangle)_{M(i)} \quad (7)$$

where $b = 1$ if $\mu(i) = 0$ and $b = -1$ if $\mu(i) = 1$. To align with Kerckhoff’s principle, the security of the system is strengthened by ensuring that K_{TA} remains private and dynamically updated. Only Alice and Trent know this coding rule,

TABLE 2 | The process of generating the quantum states from blinded message.

$\mu(i)$	K_{TA}	$M(i)$	$ \phi\rangle_{M(i)}$
0	0	1	$\frac{1}{\sqrt{2}}(0\rangle + 1\rangle)_{M(1)}$
	1	0	$\frac{1}{\sqrt{2}}(0\rangle - 1\rangle)_{M(0)}$
1	0	1	$\frac{1}{\sqrt{2}}(0\rangle + 1\rangle)_{M(1)}$
	1	0	$\frac{1}{\sqrt{2}}(0\rangle - 1\rangle)_{M(0)}$

keeping the message hidden from others. Table 2 shows the process of generating the quantum states from the blinded message.

Example: Blinding Quantum States

Let's assume $\mu(1) = 0$, and Alice selects $K_{TA} = 0$. Using Equation (6), the blinded quantum state will be:

$$|\phi\rangle_{M(1)} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

If $\mu(2) = 1$ and $K_{TA} = 1$, the state becomes:

$$|\phi\rangle_{M(2)} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

4.3 | Authorizing and Signing Phase

- After Alice agrees for Charlie to sign message on her behalf, the latter helps in the implementation of controlled QT. Alice conducts B.M.'s on the particles (μ , 1) and logs the measurements as

$$\Omega_A = \{a(1), a(2), \dots, a(l)\} \quad (8)$$

The measurements Ω_A are encrypted with $E_{K_{AB}}$ and $E_{K_{TA}}$ to obtain $E_{K_{AB}}\{\Omega_A\}$ and $E_{K_{TA}}\{\Omega_A\}$, respectively.

- Through the quantum channel, Alice sends $E_{K_{AB}}\{\Omega_A\}$ and $E_{K_{TA}}\{\Omega_A\}$ to Bob and Trent, respectively.
- Bob upon receiving the information $E_{K_{AB}}\{\Omega_A\}$, decrypts it with K_{AB} to obtain the output Ω_A . Thereafter, he encrypts (See in Example 3) Ω_A with the key K_{BC} to produce $E_{K_{CB}}\{\Omega_A\}$.

Example 3. Once Alice measures her particles and obtains the classical result Ω_A , she encrypts this classical information using the shared secret key K_{AB} obtained from the QKD protocol. Similarly, the measurement result Ω_C from Charlie can be encrypted using the key K_{CB} , securing the classical data while the quantum channel ensures the secure transmission of quantum states. Quantum states are transmitted over a quantum-secured channel. Eavesdropping detection techniques, such as the random sampling of qubits, ensure that the quantum channel remains secure. The states themselves are not encrypted but are protected by the physical principles of quantum mechanics.

- Charlie receives the encrypted state $E_{K_{CB}}\{\Omega_A\}$ and decrypts it with the key K_{CB} to get Ω_A . Charlie performs a Hadamard

operation on his particle 3 and finally performs single particle measurements based on $\{|0\rangle, |1\rangle\}$ and logs the output as

$$\Omega_C = \{c(1), c(2), \dots, c(l)\} \quad (9)$$

- Charlie encrypts $\{\Omega_A, \Omega_C\}$ using K_{CB} to generate the signature;

$$S = E_{K_{CB}}\{\Omega_A, \Omega_C\} \quad (10)$$

He, then, sends the signature S to Bob as proxy authorization.

- Charlie sends $E_{K_{TC}}\{\Omega_A, \Omega_C\}$ to Trent by using secret key K_{TC} .

4.4 | Verifying Phases

- Bob, upon receiving the S , decrypts the sign using K_{BC} and retrieves the measurement data Ω_A and Ω_C . If Ω_A values extracted from the signature do not match with the Ω_A values shared earlier by Alice, verification of the signature will be stopped, and we will call the signature invalid. While the values of Ω_A match, we continue with the following steps:
- Bob performs the required operation on particles (2) to extract the state $|\phi\rangle_{M'}$, containing information about the message.
- To counter potential forgery attacks, Trent cross-verifies Bob's decoded message M' against an encrypted ledger of valid signatures.
- Bob forwards the quantum states $|\phi\rangle_{M'(i)}$ to Trent. Trent measures them and retrieves $M'(1), M'(2), \dots, M'(l)$. He unblinds $M'(i)$ using K_{TA} , reconstructing the message μ' as

$$\mu' = K_{TA} \oplus M', i = 1, \dots, l \quad (11)$$

- Trent compares μ' with the original message μ . If they match, the protocol is deemed successful; otherwise, it fails.

Figure 1 shows the communication process in different phases and entities. Also, Figure 2 shows the flow of the proposed protocol.

4.5 | Correctness Proof

The correctness of the proposed Quantum Blind Proxy Signature (QBPS) scheme is ensured by the entanglement properties of the GHZ state. Alice performs a Bell measurement on entangled particles $\{\mu, 1\}$, ensuring that quantum correlations are preserved. Alice encrypts and transmits the measurement results Ω_A to Charlie and Bob using a quantum-secured channel. Charlie measures his particle (3) in the $\{|0\rangle, |1\rangle\}$ basis and encrypts Ω_C before sending it to Bob. Bob verifies entanglement consistency by ensuring that Ω_A and Ω_C match the expected GHZ correlations. Bob reconstructs M and unblinds it using XOR with the secret key K_{TA} :

$$M(i) = K_{TA} \oplus \mu(i)$$

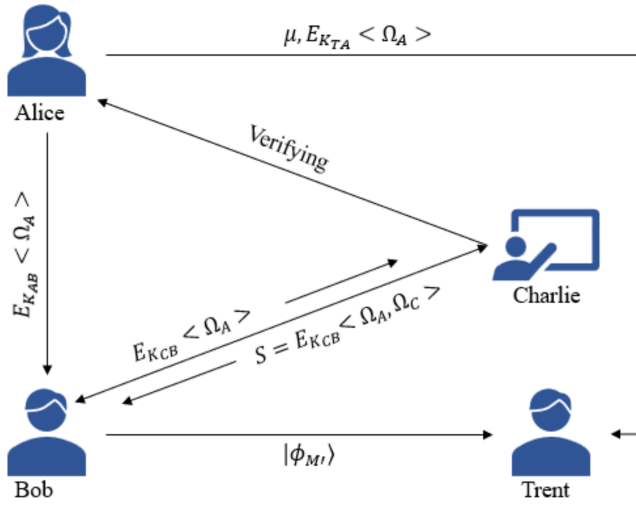


FIGURE 1 | Communication process in different phases and entities.

If Bob correctly reconstructs the message μ' , the scheme maintains correctness:

$$\begin{aligned}\mu'(i) &= K_{TA} \oplus M'(i) \\ \mu'(i) &= K_{TA} \oplus K_{TA} \oplus \mu(i)\end{aligned}$$

Since XOR follows the property $K_{TA} \oplus K_{TA} = 0$, we obtain $\mu'(i) = \mu(i)$. Therefore, if $M'(i) = M(i)$, the protocol ensures message integrity, confirming that the blind signature remains valid.

5 | Security Analysis

We are going to demonstrate that our scheme complies with the following safety criteria in this section:

- **Unforgeability:** We have assumed Trent to be a trusted participant. Due to the presence of a trusted participant, we can easily prove Charlie's ability to forge signatures (Alice is the original owner of the signature) null and void. Trent is capable of stopping such attacks within the structure of our scheme. In case Charlie tries to forge the signature of Alice, even if he possesses the key, his attempts at forgery will definitely be identified by Trent. Additionally, the availability of measurements at Trent will render any attempt by Bob to forge the signature invalid. Therefore, we can conclude that internal attackers are incapable of fabricating signatures. Second, we make the assumption that an external attacker, Eve, exists. However, Eve is not able to obtain the key shared through QKD due to the unconditional security making her incapable of forging their signatures. Even if there is a possibility that Eve randomly obtains the security key and generates a similar signature, the probability approaches zero when the bit sequence tends to infinity. We come to the conclusion that signing can only happen over a secure channel; otherwise, the process terminates immediately and the attempt to forge the signature is thwarted.
- **Undeniability** The participants cannot deny their activity in a secure quantum signature scheme, means that signer

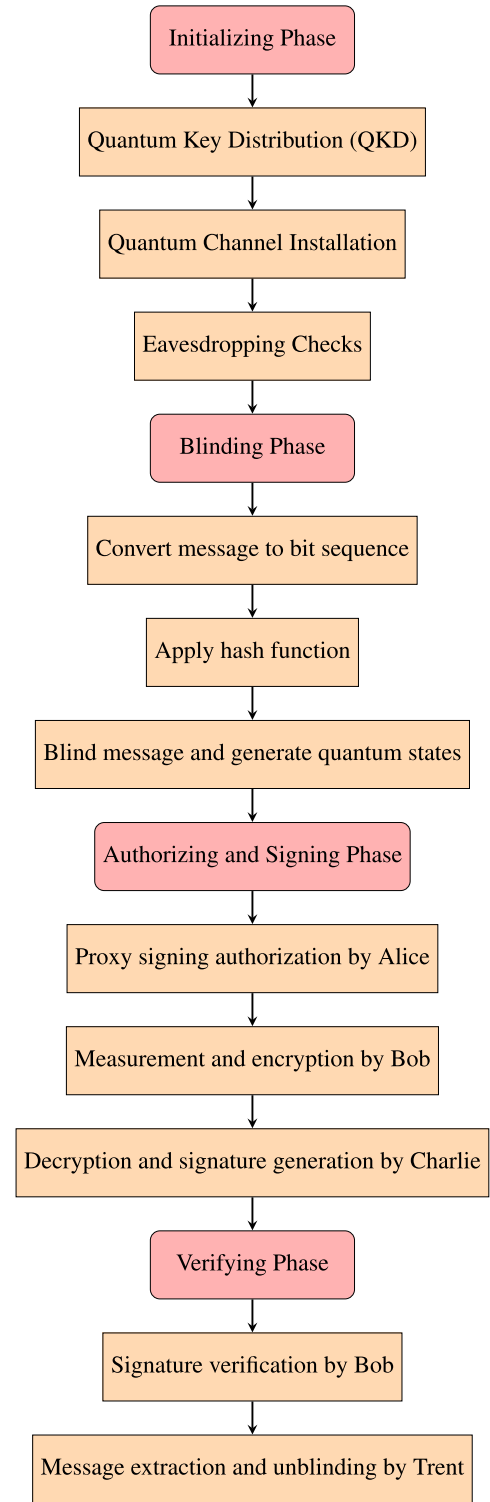


FIGURE 2 | Flow of proposed protocol.

and receiver cannot claim they did not sign and receive the signature, respectively.

1. *Disavowal of the signature is not possible for the proxy signer, Charlie.* We have verified that Alice's proxy authorization cannot be declined, and Charlie's signature cannot be declined either. If the signature is verified, the person signing it is incapable of declining either the

signature or the accompanying message. These circumstances arise due to a set of factors, which are as follows:

- a. Secret key is used to share the signature.
- b. Their is stable coherence between the entangled states between particles.
- c. Quantum channel is used to send the message.

This makes Alice unable to deny that she authorized Charlie, and impossible for Charlie to refuse his signature.

2. *Disavowal of the signature is not possible for the verifier, Bob.* If Charlie modifies the signature S to S^* , his activity will be detected when Bob compares Ω_A and Ω_C with Ω_A . In this particular scheme, the detection of eavesdropping serves as a means of identity authentication, thereby making it difficult for both Alice and Bob to deny their activity.

Since we are using QKD (quantum entanglement) for processing our QT, access to this system is exclusive to authorized participants. QKD is used to send messages from one participant to another, and only those two have access to that specific key. So, the receiving party can confirm the validity of the sending party and vice versa. Also, the properties of entangled particles make repudiation of any type by the users impractical.

- **Blindness:** In this protocol, the blindness property is achieved by Alice blinding the message μ using a hash function and XOR operation. Alice applies a hash function $h(\mu)$ and blinds the message as $M(i) = K_{TA} \oplus \mu(i)$, where K_{TA} is the shared key between Alice and Trent. This ensures that Charlie, the proxy signer, cannot decipher the original message μ . Even though Charlie creates a signature, he is unaware of the actual content of the message due to the effective blinding mechanism. This confirms the blindness property of the scheme.

- **Verifiability:**

1. *Bob's Verification of the Signature:* After receiving Alice's measurements on particle 1 and Charlie's measurements on particle 3, Bob compares the two sets of results. By matching these measurements, Bob is able to verify the authenticity of the proxy signature. The consistency between Alice's and Charlie's measurement outcomes ensures that the signature has been correctly generated by the authorized proxy (Charlie), thereby validating its authenticity.
2. *Trent's Verification of the Message:* To ensure the correctness of the message, Trent performs his own verification. Alice has already provided the original message to Trent beforehand. When Bob shares the blinded version of the message with Trent, Trent decodes the blinded message using the appropriate key. After decoding, Trent retrieves the original message and compares it with the one Alice sent earlier. If both messages are identical, Trent confirms the message's authenticity. This cross-verification between the original and decoded messages ensures that the integrity of the message has been maintained throughout the process.

- **Identifiability:**

Bob, as the signature verifier, can reliably trace the signature due to the inherent properties of quantum communication and the use of entangled particles. The entangled states enable Bob to maintain a secure and traceable link with both Alice and Charlie throughout the signature process. In the event of any dispute, Bob can utilize the quantum channel to retrieve the signature data and verify its authenticity. This traceability ensures that the signature can be unequivocally linked to the parties involved, providing a robust mechanism for resolving any conflicts that may arise. Thus, the design of our protocol guarantees a high level of identifiability for the signatures exchanged.

- **Nonrepudiation** Since we are using QKD (quantum entanglement) for processing our QT, access to this system is exclusive to authorized participants. QKD is used to send messages from one participant to another, and only those two have access to that specific key. So, the receiving party can confirm the validity of the sending party and vice versa. Also, the properties of entangled particles make repudiation of any type by the users impractical.

- **Traceability:** Bob, the signature verifier, can always trace the signature since the communication is through entangled particles and a quantum channel. So, even if any kind of dispute arises, Bob would be able to trace the signature.

- **Intercept and resend attack:** If the eavesdropper is able to intercept the qubits transferred by the sender to the recipient, Eavesdroppers transfer a series of counterfeit qubits to Bob. The sender and receiver individually execute operations of entanglement on qubits and detection particles while authenticating the channel. If Bob receives the counterfeit qubits, the sender and receiver will obtain a false detection particle or face difficulty in segregating its original quantum state. Consequently, the channel will not be successfully authenticated. Then, the intercept and resend attacks are declared invalid.

- **Entangle and measure attack:** If the eavesdropper seeks to obtain information by applying a suitable operation by causing an external qubit to entangle with the channel's qubits, this operation causes disruptions in the quantum channel. While authenticating the quantum channel, the receiver entwines the channel qubit and detection particle with each other while transmitting the latter to the sender. The detection particle is not recovered by the sender from the channel's quantum state due to disruptions caused by the eavesdropper. As a result, authentication of the channel fails, and hence this attack lacks validity.

- **Man in the middle attack:** Since we have blinded the message, information leaks are prevented since the proxy signer cannot read the message due to encryption. Signatures can also not be forged by internal or external attackers in our scheme, and the entanglement is verified through entanglement checking mechanisms. So, any attackers can neither intercept nor modify the messages between participants.

6 | Performance Analysis

Quantum mechanics is the fundamental theory that describes the behavior of particles at the quantum level. Quantum communication relies on the principles of quantum mechanics to secure the transmission of information. Simulating quantum signatures allows researchers to better understand the behavior of quantum systems, including phenomena such as superposition, entanglement, and quantum interference. It also helps to design and optimize quantum communication protocols, ensuring the security and efficiency of quantum communication systems. It plays a key role in optimizing quantum technologies, such as sensors and imaging devices. Understanding the quantum behavior of particles in these applications allows researchers to enhance the sensitivity, precision, and capabilities of these technologies. Our communication protocol employs quantum encoding, which applies the fundamental concepts of quantum physics to improve security and render it resistant to attacks from conventional computers. By doing this, the protocol's security is enhanced from "computational security" to "unconditional security" in information theory, boosting its countermeasures against potential threats. In addition, the protocol enhances overall security by diminishing dependence on possibly unreliable entities through the reduction of third-party trustworthiness.

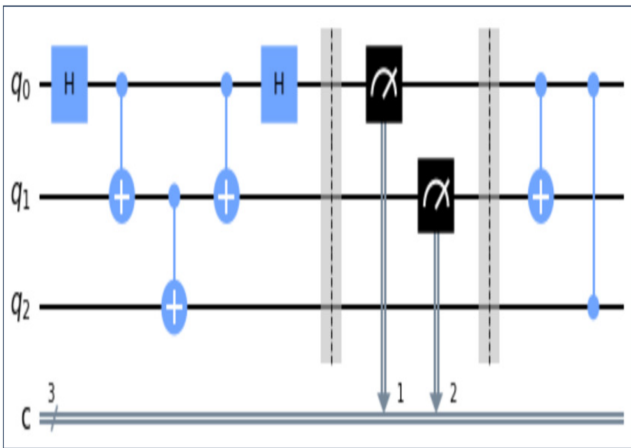


FIGURE 3 | Quantum circuit diagram.

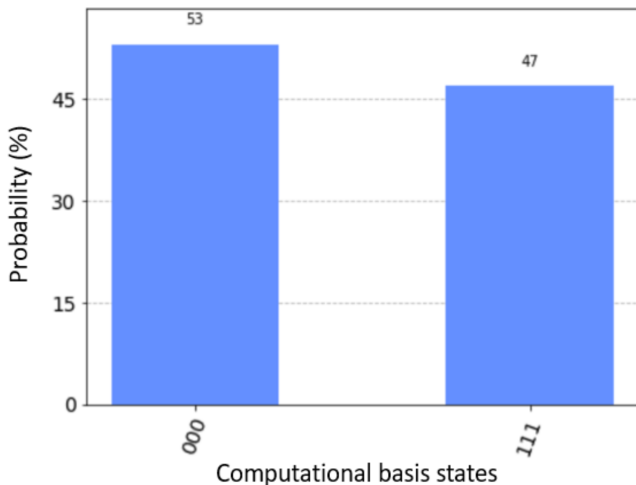


FIGURE 4 | Theoretical results.

6.1 | Used Environment

The simulation environment under consideration has the following factors:

- **Hardware environment:** We conducted experiments on a machine using 11th Gen Intel(R) Core(TM) i7-1165G7 laptop @ 2.80GHz processor.
- **Software environment:** We utilize Python 3.8.11 for coding, employing GMP, and compiling with optimization options. We also employ the "Qiskit" and "pylatexenc" packages to carry out quantum simulations with suitable parameters. The "AerSimulator" backend functions by emulating the operation of an actual device. Performing a quantum circuit with measurements will result in the return of a "count dictionary" that holds the final values of the circuit's classical registers. The circuit might have a unique instruction set, documented in a separate notebook, which includes measurements, gates, conditionals, resets, and other components.

6.2 | Experiment Analysis

We choose for single photon measurements rather than joint measurements due to the limitations of quantum devices. Figure 3 shows the circuit design for the experiment, which was carried out on the IBM-Perth quantum computer.

Using the $|\phi\rangle_{ij}$ (where $i, j \in \{0, 1\}$) state as the beginning state, Alice selects the $|\phi\rangle_{00}$, Bob selects the $|\phi\rangle_{01}$, Charlie selects the $|\phi\rangle_{10}$, Trent selects the $|\phi\rangle_{10}$ and the potential outcomes are shown in Figure 4.

The experimental results obtained from running the IBM-Perth quantum computer 5000 times are shown in Figure 5.

The experiment demonstrates a 93.33% accuracy rate, attributing any discrepancies to errors in the quantum device. Furthermore, the experimental outcomes align with the theoretical predictions, affirming that our protocol enables a just exchange of quantum information.

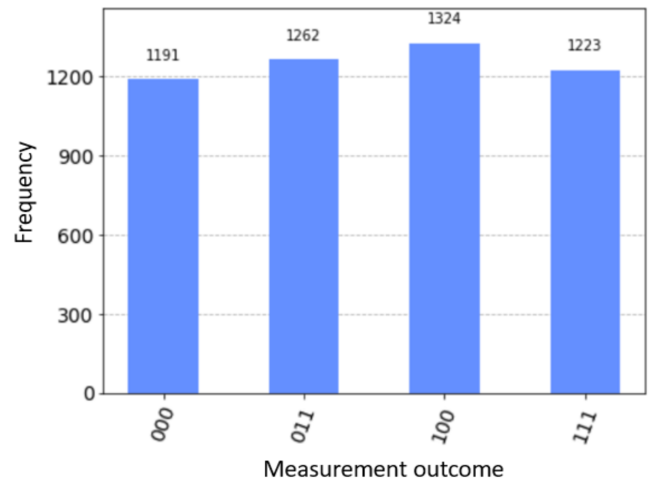


FIGURE 5 | Experimental results.

Scyther results : autoverify							
Claim				Status		Comme	
Satellite	A	Satellite,A1	Secret aj	Ok	Verified	No attacks.	
		Satellite,A2	Secret SK	Ok	Verified	No attacks.	
		Satellite,A3	Secret Mj	Ok	Verified	No attacks.	
		Satellite,A4	Alive	Ok	Verified	No attacks.	
		Satellite,A5	Weakagree	Ok	Verified	No attacks.	
		Satellite,A6	Niagree	Ok	Verified	No attacks.	
		Satellite,A7	Nisynch	Ok	Verified	No attacks.	
	C	Satellite,C1	Secret Mj	Ok	Verified	No attacks.	
		Satellite,C2	Secret SK	Ok	Verified	No attacks.	
		Satellite,C3	Secret aj	Ok	Verified	No attacks.	
		Satellite,C4	Alive	Ok	Verified	No attacks.	
		Satellite,C5	Weakagree	Ok	Verified	No attacks.	
		Satellite,C6	Niagree	Ok	Verified	No attacks.	
		Satellite,C7	Nisynch	Ok	Verified	No attacks.	
	B	Satellite,B1	Secret aj	Ok	Verified	No attacks.	
		Satellite,B2	Secret SK	Ok	Verified	No attacks.	
		Satellite,B3	Secret Mj	Ok	Verified	No attacks.	
		Satellite,B4	Alive	Ok	Verified	No attacks.	
		Satellite,B5	Weakagree	Ok	Verified	No attacks.	
		Satellite,B6	Niagree	Ok	Verified	No attacks.	
		Satellite,B7	Nisynch	Ok	Verified	No attacks.	
Done.							

FIGURE 6 | The results of the Scyther tool for the suggested procedure are shown.

6.3 | Efficiency Analysis

The efficiency of the proposed protocol is determined using the widely accepted definition given in [50]:

$$\eta = \frac{c}{u_q + u_b} \quad (12)$$

where

- c is the number of usable particles that contribute to the final message transmission,
- u_q represents the total number of qubits utilized in the protocol,
- u_b represents the total number of classical bits required.

In our protocol, the relevant contributions to c , u_q , and u_b are:

1. **Usable Particles (c):** In the quantum proxy blind signature process, the usable particles correspond to those that

successfully transmit the quantum states required for signature verification and message retrieval. Each GHZ state entanglement process involves multiple parties, but ultimately, the useful quantum states that carry verifiable information form the core of our calculation.

2. **Total Utilized Qubits (u_q):** The protocol employs GHZ states, quantum key distribution, quantum measurements, and state transmission, all of which contribute to the overall qubit consumption.
3. **Total Utilized Classical Bits (u_b):** The classical bit usage accounts for encrypted communications, verification procedures, and protocol synchronization.

Given our protocol structure, we revise the efficiency calculation as follows:

$$\eta = \frac{2(n+1) + 3 \times 2(n+1) + 3 \times 2(n+1)}{2(n+1) + 3 \times 2(n+1) + 4 \times 2(n+1)} \quad (13)$$

TABLE 3 | Comparisons of security features.

Scheme	[55]	[44]	[56]	[57]	[58]	[59]	[60]	Proposed
Message space	Quantum	Quantum	Quantum	Classical	Classical	Quantum	Classical	Quantum
Robust blindness	×	✓	✓	✓	×	✓	×	✓
Signature Space	Quantum	Quantum	Quantum	Quantum	Quantum	Quantum	Quantum	Quantum
Proxy signer authentication	×	✓	×	✓	✓	✓	✓	✓
Quantum resource	Entangled states	Entangled states	Bell states	Bell states	Bell states	Entangled states	Entangled states	GHZ-type states
Undeniability	×	×	×	✓	✓	×	×	✓
Number of protocol participants	3	3	2	2	3	4	3	4
Non-redundant proxy signing	✓	✓	✓	×	✓	✓	×	×
Dynamic proxy signing capability	✓	×	×	✓	✓	×	✓	✓

Expanding the numerator and denominator:

$$\eta = \frac{2(n+l) + 6(n+l) + 6(n+l)}{2(n+l) + 6(n+l) + 8(n+l)} \quad (14)$$

$$\eta = \frac{14(n+l)}{16(n+l)} \quad (15)$$

$$\eta = \frac{14}{16} = 0.875 = 87.5\% \quad (16)$$

6.4 | Scyther Tool

The paper outlines the formal security verification and simulation of the framework utilizing the Scyther simulation tool [25, 51–54], an efficient instrument for evaluating and detecting potential vulnerabilities and threats in network security protocols. Utilizing the Scyther tool, we illustrate the method's robustness against diverse cryptographic assaults. Scyther employs a security protocol specification as input to delineate security attributes, termed claim events and outputs, in the overview report and graph for each distinct category of cryptographic assault. The framework is delineated in the "Security Protocol Description Language (SPDL)." Moreover, the protocol specification delineates the specific sequence in which the Alice, Charlie, and Bob participate. Figure 6 illustrates the results of the security verification of the proposed protocol utilizing the Scyther tool. The results indicate that none of the claims in the proposed protocol were vulnerable to cryptographic attacks.

6.5 | Security Features

Table 3 presents an evaluation of the suggested framework based on several safety features and certain relevant protocols. The following comparative parameters are taken into consideration: a) Message space, b) Robust blindness, c) Signature Space, d) Proxy signer authentication, e) Quantum resource, f) Particle

re-usability g) Number of protocol participants, h) Nonredundant Proxy Signing, and i) Dynamic Proxy Signing Capability. The comparison presented in Table 3 clearly illustrates the advantages that the new protocol has over the methods that have already been used.

7 | Conclusion

In this article, we propose a quantum secure proxy blind signature scheme to process QT. Our signature possesses the properties of a proxy and a blind signature based on a three-particle GHZ-entangled state. Quantum key distribution, entanglement correlation, and XOR operations are employed to enhance security. This scheme has foreseeable applications in various fields, including e-payment, e-shopping, e-voting, e-banking, and e-business. Our scheme satisfies security properties such as unforgeability, nonrepudiation, and verifiability. This scheme is resilient to different security attacks, including intercept and resend attacks, entangle and measure attacks, and man-in-the-middle attacks. Therefore, our scheme is reliable enough to be applied in specific environments.

Author Contributions

Sunil Prajapat: conceptualization, software, validation, methodology, writing – original draft. **Mohammad S. Obaidat:** review, validation. **Vivek Bharmak:** methodology, writing – original draft. **Garima Thakur:** methodology, writing. **Pankaj Kumar:** supervision, review, validation.

Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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